Crystal Hoyt — *Curriculum Vitae*

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Current Address

Department of Mathematics
The Weizmann Institute of Science
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Education

· **University of California, Berkeley**
Ph.D., Mathematics, May 2007.
Dissertation: "Kac-Moody superalgebras of finite growth".
Advisor: Vera Serganova.

· **Texas A & M University, College Station**
B.S., Mathematics, May 2001, summa cum laude.

Employment

Weizmann Institute of Science, Israel
Postdoctoral Fellow, July 2007 - current.

Research Interests

Representation theory, Infinite dimensional algebras, Lie (super)algebras, W-algebras.

Honors and Awards

· 2004 Mentored Research Award, UC Berkeley Graduate Division.

· Honorable mention for 2002 NSF Graduate Research Fellowship.

· Honorable mention for 2001 Alice T. Schafer Prize, AWM.

Publications

· C. Hoyt and V. Serganova, *Classification of finite-growth general Kac-Moody superalgebras*, Communications in Algebra **35 no. 3** (2007), 851-874.

· C. Hoyt and S. Reif, *Simplicity of vacuum modules over affine Lie superalgebras*, http://arxiv.org/abs/0806.2605, to appear in the Journal of Algebra.

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Preprints

C. Hoyt, *Regular Kac-Moody superalgebras and integrable highest weight modules*, http://arxiv.org/abs/0810.2650.

Conference Talks

· *Finite-growth general Kac-Moody superalgebras*, Workshop on Representation Theory of Lie Algebraic Systems, The Weizmann Institute of Science, Rehovot, Israel, December 2006.

· *Regular Kac-Moody superalgebras*, American Mathematical Society Western Section Meeting, San Francisco State University, April 2006.

· *Classification of finite-growth Kac-Moody superalgebras*, Lie Theory Workshop, Mathematical Sciences Research Institute, Berkeley, CA, January 2005.

· *Lie algebras and Lie groups (expository)*, Mentoring Program for Women in Mathematics, Institute for Advanced Study, Princeton, NJ, May 2001.

· *Irreducible representations of solvable Lie superalgebras*, Nebraska Conference for Undergraduate Women in Mathematics, University of Nebraska - Lincoln, February 2001.

Seminar Talks

· *Simplicity of vacuum modules over affine Lie superalgebras*, Nara Women's University, Japan, November 2008.

· *Regular Kac-Moody superalgebras and integrable highest weight modules*, Nara Women's University, Japan, November 2008.

· *Representation theory of Lie superalgebras (survey)*, Nara Women's University, Japan, November 2008.

· *Contragredient Lie superalgebras of finite growth*, The Weizmann Institute of Science, Israel, June 2006.

· *An introduction to Lie algebras and representations (expository)*, University of California, Berkeley, October 2005.

· *Schur-Weyl duality and Howe duality for Lie superalgebras (expository)*, University of California, Berkeley, April 2004.

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Teaching

· Lecturer, Weizmann Institute of Science

In the winter of 2007-2008, I taught the graduate course "Lie algebras". I prepared my own material, typed my lectures, and assigned weekly homework (see my website for course materials). My evaluation scores for this course were 4.3, 4.2 out of 5.0, for course quality and instructor quality, respectively. In the summer of 2007, I taught a minicourse on Lie algebras for visiting summer students. I am currently giving a reading course on Lie algebras for a masters student.

· *Graduate Student Instructor, University of California, Berkeley*

In the summer of 2004 and 2005, I was fully responsible for teaching Math 16A: Analytic Geometry and Calculus. This included developing my course syllabus, assigning homework, writing and grading quizzes and exams, running a discussion section, and lecturing for the course. My evaluation scores for this course were 6.0, 6.3 out of 7.0 in Summer 2004, and 5.9, 6.2 out of 7.0 in Summer 2005, for course quality and instructor quality, respectively. I have also worked as a teaching assistant, Fall 2002 - Spring 2004, where I ran discussion sections and graded.

· *Residential Counselor, SUMaC, Stanford University*

The Stanford University Math Camp exposes talented high school students to advanced topics in math by using both the standard classroom approach and the research motivated style of teaching. In the summer of 2003, I worked as a SUMaC residential counselor, where I led a group research project on coding theory, I organized various activities for the students, and I supervised and tutored the students.

· *Math Tutor for Help Sessions, Texas A & M University*

As an undergraduate, I was employed by the university to work in "Help Sessions" as a math tutor for Linear Algebra and Calculus, Spring 1999 - Spring 2001. These are scheduled tutor sessions, occurring several times per week for each course, that are held in empty classrooms in the evening. The students are invited to sit and work on their homework, and they are able to ask questions when they need help. I would walk around the classroom to answer questions, and discuss more difficult concepts on the board. My goal was to teach them how to solve problems by themselves.